



Sri Sai Vidya Vikas Shikshana Samithi®

SAI VIDYA INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi, affiliated to VTU, Recognized by Govt. of Karnataka.

Accredited by NBA, New Delhi (CSE, ECE, ISE), NAAC – “A” Grade

DEPARTMENT OF Computer SCIENCE AND ENGINEERING

RAJANUKUNTE, BENGALURU – 560 064, KARNATAKA

Phone: 080-28468191/96/97/98 * E-mail: hoddatscience@saividya.ac.in * URL www.saividya.ac.in



Module-1: Big Data Analytics	BTL	CO
1. What is data? Explain the characteristics of data.	2	1
2. Explain the classification of data.	2	1
3. What is big data? How is big data generated?	1	1
4. Explain the importance of big data.	2	1
5. Explain the challenges of big data.	2	1
6. Differentiate between Traditional Business Intelligence Vs Big Data	2	1
7. Write a short note on typical data warehouse and typical hadoop environment.	2	1
8. What is Big data analytics? Explain the classification of Big data analytics.	2	1
9. Write a short note on the following: • Descriptive analytics • Diagnostic analytics • Predictive analytics • Prescriptive analytics	2	1
10. Why Big data Analytics is important?	2	1
11. Explain the technologies used in Big data Environments.	2	1
12. Explain CAP Theorem.	2	1
13. Write a short note on NoSQL. (Explain the features, what it is, where it is used, types and advantages)	2	1
14. Difference between SQL and NoSQL	2	1
15. What is NewSQL? Explain the Characteristics of NewSQL	2	1
16. Comparison of SQL, NoSQL, and NewSQL	2	1
17. What is Hadoop? Explain the features and advantages of hadoop.	2	1
18. Explain the versions of Hadoop.	2	1
19. Explain the Hadoop ecosystem components for Data ingestion, data processing and data analysis.	2	1
20. Difference between HBase and Hadoop/HDFS	2	1
21. Difference between Hive and RDBMS	2	1
22. Difference between Hive and HBase	2	1
23. Explain about Hadoop distributions.	2	1
24. Difference between Hadoop v/s SQL	2	1

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Module-2: Introduction to Hadoop	BTL	CO
1. What is Hadoop? What is the need of Hadoop?	1	2
2. Why Hadoop? Why not RDBMS?	1	2
3. Differentiate between RDBMS and Hadoop.	2	2
4. Explain about Hadoop overview.	2	2
5. Explain Hadoop conceptual layer and also explain the high level architecture of Hadoop.	2	2
6. Write a short note on HDFS.	2	2
7. With a neat diagram explain HDFS architecture.	2	2
8. Explain HDFS Daemons.	2	2
9. Explain anatomy of HDFS file read and anatomy of file write.	2	2
10. Write the HDFS command for the following objectives: 1. To create a directory (say, sample) in HDFS. 2. To copy a file from local file system to HDFS(say sample to test) 3. To copy a file from HDFS to local file system.(say test to test sample) 4. To display the contents of an HDFS file on console. 5. To remove a directory from HDFS.(say sample1) 6. To get the list of directories and files at the root of HDFS. 7. To get the list of complete directories and files of HDFS. 8. To copy a file from local file system to HDFS via copyFromLocal command. 9. To copy a file from Hadoop file system to local file system via copyToLocal command. 10. To copy a file from one directory to another on HDFS.	3	2
11. How does processing of data happen with Hadoop?	2	2
12. Explain Mapreduce Daemons	2	2
13. How does Mapreduce work?	2	2
14. Explain Mapreduce programming architecture.	2	2
15. Implement a word count program in Hadoop.	3	2
16. How to manage resources and applications with Hadoop YARN?	2	2
17. With a neat figure, explain Hadoop 1.0 and Hadoop 2.0 architecture. Also explain the fundamental Idea of Hadoop 2 YARN: Taking Hadoop beyond Batch.	2	2
18. Explain YARN Architecture.	2	2



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19. Explain the process of Installing Hadoop and Implement the following file management tasks in Hadoop: Adding files and directories Retrieving files Deleting files and directories.	3	2
INTRODUCTION TO MAPREDUCE PROGRAMMING		
20. Write a short note on the following with an illustrative example. 1. Mapper 2. Reducer 3. Combiner 4. Partitioner 5. Searching 6. Sorting 7. Compression Also write the Java code for the same.	3	2
21. Develop a MapReduce program to implement Matrix Multiplication	3	2
22. Develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day.	3	2
23. Develop a MapReduce program to find the tags associated with each movie by analyzing movie lens data.	3	2

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Module-3: Introduction to MongoDB	BTL	CO
1. What is MongoDB? Explain why MongoDB is required by explaining the following: a. Java Script Object Notation(JSON) b. Process of Creating or Generating a Unique Key. c. Support for Dynamic Queries d. Storing Binary Data e. Replication f. Sharding g. Updating Information In-Place	2	3
2. List the terms used in RDBMS and MongoDB with description.	1	3
3. Explain the data types in MongoDB with example.	2	3
4. Explain the following with syntax and example using MongoDB queries. a. Creating database b. Dropping database	3	3
5. Write the MongoDB shell command for the following. a. To report the name of the current database b. To display the list of all databases c. To switch to a new database d. To display the list of collections (tables) in the current database e. To display the current version of the MongoDB server	3	3
6. Given a collection named Students with the following fields: StudRollNo, StudName, Grade, Hobbies, DOJ Write MongoDB commands to perform the following operations i) Insert the details of one student ii) Display the details of the student having rollno 101 iii) Change the hobby of the student named “Rahul” from Cricket to Football iv) Delete the record of the students belonging to V grade. v) Retrieve the details of students whose names start with the letter 'B'	3	3
7. Explain the following methods with an example for each i) Save() ii) Find() iii) Pretty()	3	3



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iv) Count() v) Skip() vi) Insert() vii) Update() viii) Remove() ix) Dealing with Null values x) Limit() xi) Sort() xii) Arrays xiii) Aggregate()		
8. Explain the following in MongoDB: a. Mapreduce function b. Java Script Programming c. Cursors in MongoDB d. Indexes e. Mongoimport f. Mongoexport g. Automatic Generation of Unique Numbers for the _id Field	2	3
9. Implement Functions: Count – Sort – Limit – Skip – Aggregate using MongoDB.	3	3

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Module-4: Introduction to Hive	BTL	CO
1. What is Hive? List the features of Hive.	1	4
2. Explain Hive Integration and Work Flow.	1	4
3. Explain Hive Data Units.	1	4
4. Explain when to use partitioning/bucketing.	2	4
5. With a neat diagram explain Hive architecture. Also explain the types of metastore.	2	4
6. Explain Hive Datatypes.	2	4
7. Explain Hive File Formats.	2	4
8. Write the commands used in Hive to do the following: a. Create a database b. To display a list of all databases available in Hive. c. To view details about an existing database in Hive. d. To describe the extended properties of a database. e. To alter the database properties. f. To make a database the current working database. g. To drop a database.	3	4
9. Explain the different kinds of table in Hive. Also write the commands for the following in Hive. a. To create a managed table named STUDENT. b. To describe the structure and metadata of the STUDENT table. c. To create external table named 'EXT_STUDENT'. d. To load data into the table from file named student.tsv. e. To work with collection data types. f. To retrieve the student details from "EXT_STUDENT" table. g. Querying Collection Data Types.	3	4
10. Write a short note on partitioning with an example of creating, loading and adding in both static partition and dynamic partition.	3	4
11. Explain bucketing with an example.	3	4
12. Explain the following with an example. a. Views b. Sub-query c. Joins d. Aggregation e. Group By and Having f. RCFILE IMPLEMENTATION g. USER-DEFINED FUNCTION (UDF)	3	4



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INTRODUCTION TO PIG		
13. What is Pig? Explain the key features of Pig.	1	4
14. With a neat diagram explain the anatomy of pig	2	4
15. Explain the use case for pig: ETL processing	2	4
16. Explain the Pig Latin statements.	2	4
17. Explain Pig Latin keywords and Identifiers	2	4
18. Explain different operators in Pig Latin.	2	4
19. Explain the data types in Pig Latin	2	4
20. How to run Pig?	2	4
21. Explain different relational operators in Pig Latin with example.	3	4
22. Explain different Eval Function in Pig Latin with example.	3	4
23. Write a short note on following with example. • Piggy Bank • User Defined Functions	3	4

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Module-5: Spark	BTL	CO
1. Explain the main features of Apache Spark.	1	5
2. Discuss the role of Spark SQL and its components.	2	5
3. Describe the Spark Software Stack.	2	5
4. Write a note on the process of Data Analysis using Spark.	2	5
5. Explain the steps involved in the ETL process.	2	5
6. How can DataFrames be created using JSON and Parquet formats?	2	5
7. Write a short note on SparkR and Spark MLlib.	2	5
8. Explain the usage of Python libraries like NumPy and SciPy in data analysis.	3	5
9. Describe the process to register and use UDFs in Spark SQL.	2	5
10. Compare regular UDFs and vectorized UDFs with examples.	2	5
11. Describe the Spark SQL architecture with a diagram. Explain how it connects with Hive, JSON, and Parquet.	2	5
12. Elaborate on the Spark stack layers and explain each with suitable examples.	2	5
13. Discuss in detail the various steps in performing data analysis with Spark including preprocessing and transformation.	2	5
14. How does Python integrate with Spark for advanced data analytics? Illustrate with examples using PySpark and UDFs.	3	5
15. Write a detailed note on Grouped Vectorized UDFs. Include syntax, steps, and use cases.	2	5
16. With examples, explain how to perform join operations and aggregation in Spark SQL using DataFrames.	3	5
17. Explain the use of NumPy, SciPy, and Pandas in the context of big data analysis.	3	5
18. Describe the process of outlier detection and removal using Python in Spark-based data analysis.	3	5

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